Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 1 of 14 · Question 3 part a · 2 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

Question number				
3	(a)	Any two from: • diseases caused by pathogens/bacteria/viruses; • diseases can be transferred from one person to another; • diabetes caused by failure to produce insulin/not caused by pathogen;		2
	(b) (i)	• 25%/1 in 4 don't know; • $\frac{25 \times 5\,000\,000;}{100}$ OR $\frac{5\,000\,000;}{4}$ • = 1.25 (million);	ecf max 1 mark for correct final answer to a calculation	3
	(ii)	• $\frac{10 \times 5\,000\,000;}{100}$ • = 0.5 (million);	ecf max 1 mark for wrong calculation but correct answer to the calculation	2
	(c) (i)	glucose;		1
	(ii)	Any two from: • reference to change in the level of fluid in the eye/ear/leakage of blood into the eye; • blood vessels/neurones in ear/cochlea/eye/retina may be damaged; • reduced blood flow to eye/ear/less nutrients/oxygen delivered;		2
	(iii)	Any three from: • glucose/sugar in filtrate/urine; • water moves from cells/tissues into filtrate/urine; • reference to water potential in filtrate/kidney being lower than in surrounding tissue/less water reabsorbed;	Allow reference to water moving from high to low water potential	3
				Total 13

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 1 of 14 · Question 3 part a · 2 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 2 of 14 · Question 4 part b · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

4 (a)	$\frac{30\,000}{100}$ x 40; 12 000;	Full marks for final correct answer ecf	1 1
(b)	a microorganism/named microorganism that causes disease;		1
(c)	It cannot be transmitted from one person to the next		1
(d)	• use a sample of people/use two groups of people; • feed each sample/group of people a different diet/feed one group a diet with vitamin B and the other without vitamin B; • monitor health/compare (health) of two groups;		1 1 1
(e)	• breakdown of cartilage/cartilage not formed properly/damaged; • bones rub together/more friction between bones/less shock absorption;		1 1

Total question 4 = 9 marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 2 of 14 · Question 4 part b · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 3 of 14 · Question 6 part c.i · 1 mark

2020 JAN · Paper 01 · find part c.i in this mark scheme extract

Number				
6	(a)	bronchus/bronchi (1)		1
	(b) (i)	• inhale and exhale/breathe in an out (1) • through mouthpiece (1) • floating air chamber rises/falls (1) • pen records a trace on (the rotating) graph paper/drum (1)		4
	(ii)	Do not share the mouthpiece with another person/disinfect mouthpiece before use by another person/after use (1)		1
	(iii)	Any two from: • airways /bronchi/bronchus constricted/obstructed/reduced diameter (1) • less/reduced airflow (1) • per unit time (1)		2
	(c) (i)	cannot be transmitted from one person to another/not contagious/not caused by a pathogen/bacteria/virus/microorganism (1)		1
	(ii)	9.5 - 7 = 2.5(1) 2.5 ÷ 7 x 100(1) 36(%) (1)	Full marks for correct final answer Allow 35.7 for final answer	3
	(iii)	people taking less exercise/ consuming too much fat/sugar/ taking in more energy than is being used/ increase in the number of people smoking/drinking more alcohol/obesity (1)	Allow other valid lifestyle factor	1

Total for Question 6 = 13 marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 4 of 14 · Question 7 part a.i · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number				
7				
(a)	(i)	C; (pathogen) A not all bacteria cause disease B not all fungi cause disease D not all protozoa cause disease		1
	(ii)	• Ebola caused by a virus; • antibiotics only affect bacteria/not active against viruses;		2
(b)	(i)	• 700 - 180 = 520; • 520 ÷ 5 (= 104); • 100 (AU per hour);	Ecf max 2 marks if first mp incorrect but value divided by 5 to arrive at correct answer Max 1 if first answer correct but division is incorrect	3
	(ii)	• B is bactericidal; • because population decreases/kills bacteria;		2
	(iii)	• antibiotic B; • because it kills bacteria/causes population to decrease; • but symptoms may be less severe/mild/reduced; • antibiotic A leaves large numbers alive/bacterial population remains constant/only stops/reduces population growth; • these can still cause disease;		3 Total 11

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 4 of 14 · Question 7 part a.i · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 5 of 14 · Question 8 part a.ii · 1 mark

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

8	(a)	(i)	A = flagellum; B = cell wall/capsule; C = nucleic acid/genetic material/bacterial chromosome/(loop of) DNA;		3 (grad)
		(ii)	pathogen;		1 (cler)
	(b)	(i)	• produced by microorganism/bacteria/fungi; • kills bacteria;		2 (grad)
		(ii)	A *no mark for giving X* • because larger clear zone; • antibiotic diffused from filter paper; • bacteria have been killed;		3 (exp)
		(iii)	• filter paper without antibiotic; • bacterial growth unaffected/no clear zones; • if antibiotic is causing effect;		3 (exp)

Total 12 marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 5 of 14 · Question 8 part a.ii · 1 mark

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 6 of 14 · Question 3 part a · 1 mark

2023 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

Number				
3	(a)	pathogens;		1
	(b) (i)	1.2 years;		1
	(ii)	• active; • artificial;		2
	(iii)	Any two from • weakened/dead/attenuated organisms; • toxins; • antigens; • RNA;	Allow named bacterium/virus Ignore genetic material	2
	(c)	Any two from • passed from mother; • during breastfeeding/from breast milk; • across placenta;	Ignore umbilical cord	2
	(d)	Any five from • antibody level falls; • below safe level/to unsafe level; • greater risk of disease; • booster dose increases antibody levels/remains above unsafe level; • person immune/can't catch disease; • increased number of memory cells;	Allow faster response	5
	(e)	Any two from • antibody level higher; • for longer period/decreases more slowly/remain for many years; • levels of antibody increase faster after booster;		2

Total 15 marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 6 of 14 · Question 3 part a · 1 mark

2023 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 7 of 14 · Question 4 part a · 1 mark

2025 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

4 (a)		disease causing organism/micro-organism;		1
(b)	(i)	any three from • cytoplasm; • cell/plasma membrane; • cell wall; • (loop of) nucleic acid/DNA; • flagellum;	if more than three parts labelled minus one mark for each incorrect label	3
	(ii)	• distance between A and B = 64 mm/6.4 cm; • 64 × 1000 = 64 000 µm; • magnification = $\frac{64\,000}{2}$ = 32 000;	ecf allow +/- 1 mm	3
(c)	(i)	• suitable scale, axes labelled with units; • correct way round; • points plotted correctly; • lines joining plotted points; • key to identify lines;	needs to cover half of paper award mp1,2 and 5 if bar chart plotted	5
	(ii)	any three from • both fall/vaccine becomes less effective over time; • 45-64 years more effective (over all months)/over 65 less effective (over all months); • slowly for first two months; • more/ rapid /steep decrease, between 3-4 months/after month 3 (for over 65);		3
				15 marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 7 of 14 · Question 4 part a · 1 mark

2025 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 8 of 14 · Question 3 part a · 1 mark

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

3(a)	D; (virus)		**(1)**

Question Number	Answer	Additional guidance	Mark
3(b)	Any three from: • wear protective masks/clothes/goggles; • avoid contact with bodily fluids / avoid direct contact with infected person; • isolate/quarantine infected person; • safe burial; • disinfect surfaces to kill pathogen;		**(3)**

Question Number	Answer	Additional guidance	Mark
3(c)	Any three from: • few pathogens enter body at infection; • large numbers required to cause symptoms; • incubation; • need to reproduce/increase numbers; • time taken to do this;	Ignore reference to virus spreading	**(3 max)**

Question Number	Answer	Additional guidance	Mark
3(d)	• symptoms include severe diarrhoea / vomiting; • loss of water/salts/dehydration; • ORT replaces water/rehydrates/replaces salts;		**(3)**

Question Number	Answer	Additional guidance	Mark
3(e)	• causative pathogen found in bats/bats are vectors/carriers; • causes no ill effects/symptoms/disease;		**(2)**

Total 12 Marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 8 of 14 · Question 3 part a · 1 mark

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 9 of 14 · Question 4 part a.i · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
4	(a) (i)	A (bacterium); B doesn't cause it C doesn't cause it D doesn't cause it		1
	(ii)	• no of cases decreases as vaccination rate increases; • lag in effect; • fluctuations in number of cases; • despite percentage of vaccinations remaining constant;		4
	(b)	2.75 per 100 000; $\frac{2.75 \times 3\,450\,000}{100\,000}$; = 95 people;	correct answer without working = 3 ECF = 2	3
	(c)	• inject weakened/attenuated organism; • causes antibody production; • antibodies remain in blood; • memory cells formed; • on infection antibodies can respond quickly;		5

Total 13 marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 9 of 14 · Question 4 part a.i · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 10 of 14 · Question 7 part a · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

7 a)	bacteria;		1
(b)	• through water/food; • containing contaminated/infected faeces;	Do not need to say contaminated for mp2 if mentioned in mp1	2
(c)	Any four from the following: • oral rehydration therapy; • replacement of fluids/salts/sugar; • as water/salts are lost through vomiting / diarrhoea; • antibiotics; • to kill bacteria;		4

Total marks 7

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 10 of 14 · Question 7 part a · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 11 of 14 · Question 8 part a.i · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.i in this mark scheme extract

8 (a)	(i)	Bacteria;		(1)
	(ii)	Contaminated water/food;		(1)
(b)		• oral rehydration(therapy)/ oral rehydration (solution); • reference to electrolytes/ions/named electrolytes (mixed) in water; • Reference to clean/boiled water; • Reference to antibiotics ;		(3)

Total for Question 8 = 5 marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 12 of 14 · Question 1 part a · 3 marks

2024 JUN · Paper 02 · find part a in this mark scheme extract

1 (a)	<table><tr><th>Disease</th><th>Type of pathogen</th></tr><tr><td>cholera</td><td>Bacterium/bacteria;</td></tr><tr><td>AIDS</td><td>virus;</td></tr><tr><td>athlete's foot</td><td>Fungus/fungi;</td></tr></table>	Reject more than one line from any disease	(3)
(b)	In the following order: bacterium; sexual intercourse; barrier; antibiotics;		(4)
(c) (i)	• overall decrease; • fluctuates • steep decreases from 1990;		(2)
(ii)	17 - 9; 8;	Allow tolerance of +-0.5 ecf for mp2	(2)
(d)	C antigens,white blood cells produced,antibodies destroy pathogens; The following are incorrect because - A vaccinations do not contain antibodies B platelets are not produced following a vaccination D vaccinations do not produce platelets		(1)

Total for Question 1 = 12 marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 13 of 14 · Question 8 part b.i · 1 mark

2023 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

8	(a)	(i)	Any two from • DNA/RNA/genetic material; • surrounded by a protein coat/capsid; • (protein) spikes on outer surface;		2
		(ii)	Any five from • transcription (of DNA); • into mRNA; • translation (of mRNA) at the ribosome; • by tRNA; • reference to (tRNA) anticodon pairing with (mRNA) codon; • amino acids assembled to form (viral) proteins;		5
	(b)	(i)	Human Immunodeficiency virus/HIV;		1
		(ii)	• (1400000 – 650000) = 750000; • (750000 ÷ 1400000) (x 100) = 54%	Allow any figure that rounds to 54	2

Total for Question 8 = 10 Marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 14 of 14 · Question 1 part a · 6 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

<table>
<tr>
<td>1 (a)</td>
<td>
<table>
<tr><th>Name of disease</th><th>Type of organism</th><th>Method of transmission</th></tr>
<tr><td>malaria</td><td>protozoan;</td><td>mosquito;</td></tr>
<tr><td>poliomyelitis</td><td>virus;</td><td>air/water;</td></tr>
<tr><td>typhoid</td><td>bacterium;</td><td>food/water/housefly;</td></tr>
</table>
</td>
<td>For methods of transmission:

Allow anopheles/vector

Allow droplets for water

Allow vector for housefly</td>
<td>6</td>
</tr>
<tr>
<td>(b)</td>
<td>• caused by a fungus (1)
• (antibiotics) only effective against bacteria/ not effective against fungi (1)</td>
<td>Allow (antibiotics) do not kill fungi</td>
<td>2</td>
</tr>
</table>

Total 8 marks

Type MS · Define pathogen, classify diseases and name the organism type causing each

Answer 14 of 14 · Question 1 part a · 6 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks